

Meta-Regression

Subgroup Analysis

This vignette introduces the regression functionality of the `dtametaTMB` package for meta-analysis of diagnostic test accuracy (DTA) studies.

Validation

The subgroup HSROC and Reitsma implementations reproduce the Cochrane Handbook RF and Anti-CCP examples closely, including the baseline parameters (HSROC: accuracy, threshold, shape; Reitsma: logit sensitivity and specificity), between-study variance estimates, and subgroup effects on accuracy and threshold. The Schuetz CT/MRI analyses yielded results broadly consistent with the published subgroup parameter estimates and variance components. Likelihood-ratio tests led to identical substantive conclusions, although the LR statistics differed somewhat from the published values.

Dummy-coding vs. cell-means parameterization

For subgroup analyses, we distinguish between two equivalent parameterizations of the same model. In the reference-group parameterization, one subgroup serves as the baseline and the remaining subgroup effects are expressed as deviations from this reference. In the group-specific (cell-means) parameterization, each subgroup has its own parameter directly. The former is convenient for testing subgroup differences, whereas the latter is convenient for reporting subgroup-specific estimates and plotting subgroup-specific HSROC curves. The Reitsma subgroup model is fitted twice with both parameterizations. The subgroup HSROC model is estimated using a reference-group parameterization, and subgroup-specific parameters are recovered by evaluating the fitted linear predictors at subgroup-specific design points via Z_{pred} .

How do we include covariates in the Reitsma model?

For sensitivity in study i , we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\begin{pmatrix} z_i^\top \mu_{Ai} \\ z_i^\top \mu_{Bi} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} z_i^\top \mu_A \\ z_i^\top \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

Let's assume that z_i includes a single binary covariate representing two subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \mu_{Ai} = \begin{cases} \mu_{Ai} & \text{for subgroup 1,} \\ \mu_{Ai} + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_{Bi} = \begin{cases} \mu_{Bi} & \text{for subgroup 1,} \\ \mu_{Bi} + \nu_{B2} & \text{for subgroup 2,} \end{cases}$$

$$z_i^\top \mu_A = \begin{cases} \mu_A & \text{for subgroup 1,} \\ \mu_A + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_B = \begin{cases} \mu_B & \text{for subgroup 1,} \\ \mu_B + \nu_{B2} & \text{for subgroup 2.} \end{cases}$$

```
data("anticcp")
reitsmasub <- fitReitsmaSubgroup(data=anticcp,
                                TP=TP,
                                FP=FP,
                                FN=FN,
                                TN=TN,
                                study=study,
                                subgroup=generation)

reitsmasub
#>
#> Reitsma Subgroup Model
#> -----
#>
#> Number of studies      : 37
#> Number of subgroups   : 2
#> Optimizer              : Converged
#> Hessian                 : Positive definite
#> Max |grad|             : 0.0004844247
#> -2 log likelihood      : 533.37 ( df = 7 )
```

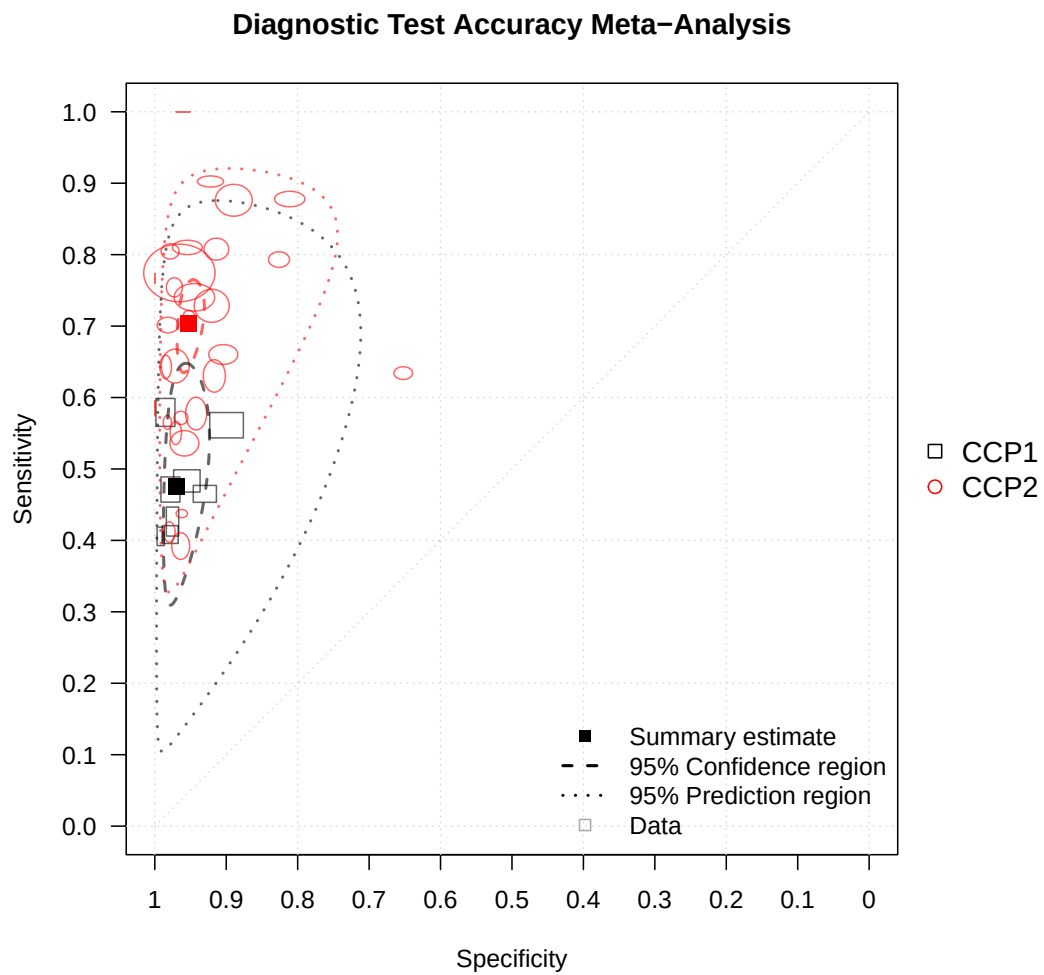
```

#> AIC : 547.37
#> BIC : 558.646
#>
#> Use summary() for parameter estimates.
summary(reitsmasub)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#> Estimate Std_Error
#> mu_A.CCP1 -0.096538884 0.220320590
#> mu_B.CCP1 3.446719200 0.298243690
#> mu_A.CCP2 0.866038480 0.120876820
#> mu_B.CCP2 3.016490600 0.162237540
#> sigma2_A.sens.log_sigma_A 0.359828710 0.102176500
#> sigma2_B.spec.log_sigma_B 0.539909310 0.180157230
#> sigma_AB -0.196844990 0.098353424
#> nu_A.CCP2 0.962571630 0.251342040
#> nu_B.CCP2 -0.430210700 0.337711820
#>
#> Sensitivity / specificity
#> -----
#>
#> type Estimate conflevel CI_Lower CI_Upper
#> mu_A.CCP1 sens 0.47588401 0.95 0.37089964 0.58304390
#> mu_B.CCP1 spec 0.96913315 0.95 0.94594449 0.98255779
#> mu_A.CCP2 sens 0.70392072 0.95 0.65229087 0.75081294
#> mu_B.CCP2 spec 0.95331358 0.95 0.93693872 0.96559260
#>
#> Recovered HSROC parameters (Rutter-Gatsonis)
#> -----
#>
#> Lambda Theta beta sigma2_alpha sigma2_theta
#> CCP1 3.0073766 -1.61053440 0.20288653 0.48784248 0.31880561
#> CCP2 3.6840001 -0.88349711 0.20288653 0.48784248 0.31880561
#>
#> Subgroups
#> -----
#> [1] "CCP1" "CCP2"

```

How do I get a summary plot of the Reitsma model?

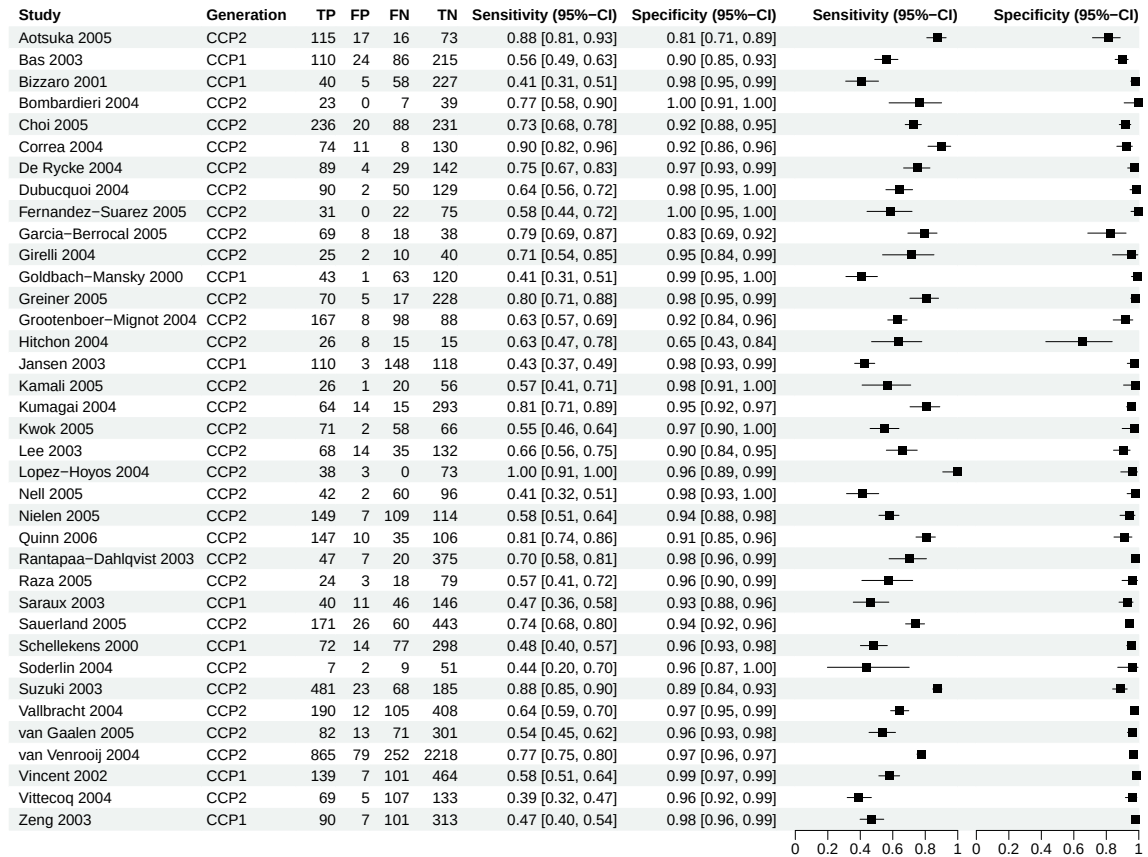
```
plot(reitsmasub,  
     scale=0.01,  
     nudge_legend=-0.2,  
     size="se",  
     col=c("black","red"))
```



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(reitsmasub, subgroup_label="Generation")
```



How do I constrain parameters in the Reitsma model?

In sparse data one may wish to fix parameters of the random effects (variance-covariance matrix) at zero. This can be done via the `constrain` argument.

For example,

```
constrainA <- fitReitsmaSubgroup(data=anticcp,
                                TP=TP,
                                FP=FP,
                                FN=FN,
                                TN=TN,
                                study=study,
                                subgroup=generation,
                                constrain="sigma2_A")

summary(constrainA)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate   Std_Error
#> mu_A.CCP1          -5.4394046e-02 0.054985303
#> mu_B.CCP1           3.4466250e+00 0.298754820
#> mu_A.CCP2           9.1797945e-01 0.031391722
#> mu_B.CCP2           2.9968587e+00 0.162162330
#> sigma2_A.sens.log_sigma_A 4.9303807e-32 0.000000000
#> sigma2_B.spec.log_sigma_B 5.3963306e-01 0.179976400
#> sigma_AB            0.0000000e+00 0.000000000
#> nu_A.CCP2           9.7237352e-01 0.063315273
#> nu_B.CCP2          -4.4976608e-01 0.337953820
#>
#> Sensitivity / specificity
#> -----
#>
#>      type   Estimate conflevel   CI_Lower   CI_Upper
#> mu_A.CCP1 sens 0.48640484      0.95 0.45954779 0.51334062
#> mu_B.CCP1 spec 0.96913033      0.95 0.94588841 0.98257334
#> mu_A.CCP2 sens 0.71463023      0.95 0.70191901 0.72701020
#> mu_B.CCP2 spec 0.95243201      0.95 0.93577764 0.96492938
#>
#> Recovered HSR0C parameters (Rutter-Gatsonis)
#> -----
#>
#>      Lambda      Theta      beta sigma2_alpha  sigma2_theta
#> CCP1 -3128642.1 -1564321.1 35.73522 3.262267e-16 8.1556674e-17
#> CCP2 52800433.0 26400216.0 35.73522 3.262267e-16 8.1556674e-17
#>
#> Subgroups
```

```
#> -----
#> [1] "CCP1" "CCP2"
```

fixes the logit sensitivity variance to zero. This also implies that the random effects covariance is zero. Note that you can also set `constrain` to `"sigma_AB"`, `"sigma2_B"`, or `"all"`.

Constraining fixed effects is controlled by the `sensspec_constrain` argument. For example,

```
constrainsens <- fitReitsmaSubgroup(data=anticcp,
                                   TP=TP,
                                   FP=FP,
                                   FN=FN,
                                   TN=TN,
                                   study=study,
                                   subgroup=generation,
                                   sensspec_constrain="sens")

summary(constrainsens)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate Std_Error
#> mu_A.CCP1          0.653305180 0.12740877
#> mu_B.CCP1          3.081222100 0.32561613
#> mu_A.CCP2          0.653305180 0.00000000
#> mu_B.CCP2          3.118005800 0.17451227
#> sigma2_A.sens.log_sigma_A 0.541939800 0.14627725
#> sigma2_B.spec.log_sigma_B 0.577648680 0.19968290
#> sigma_AB          -0.277162040 0.13981590
#> nu_A.CCP2          0.000000000 0.00000000
#> nu_B.CCP2          0.036831361 0.38577185
#>
#> Sensitivity / specificity
#> -----
#>
#>      type  Estimate conflevel  CI_Lower  CI_Upper
#> mu_A.CCP1 sens 0.65775489      0.95 0.59954955 0.71157008
#> mu_B.CCP1 spec 0.95611150      0.95 0.92004997 0.97632597
#> mu_A.CCP2 sens 0.65775489      0.95 0.59954955 0.71157008
#> mu_B.CCP2 spec 0.95762939      0.95 0.94136328 0.96952925
#>
```

```
#> Recovered HSROC parameters (Rutter-Gatsonis)
#> -----
#>           Lambda      Theta      beta sigma2_alpha sigma2_theta
#> CCP1 3.6962689 -1.1843237 0.031905471 0.56469478 0.41833574
#> CCP2 3.7324705 -1.2024245 0.031905471 0.56469478 0.41833574
#>
#> Subgroups
#> -----
#> [1] "CCP1" "CCP2"
```

assumes equal (logit) sensitivities in all subgroups. Note that you can also set `sensspec_constrain` to "spec" or `c("sens","spec")`.

How can I compare `constrainsens` with the full model?

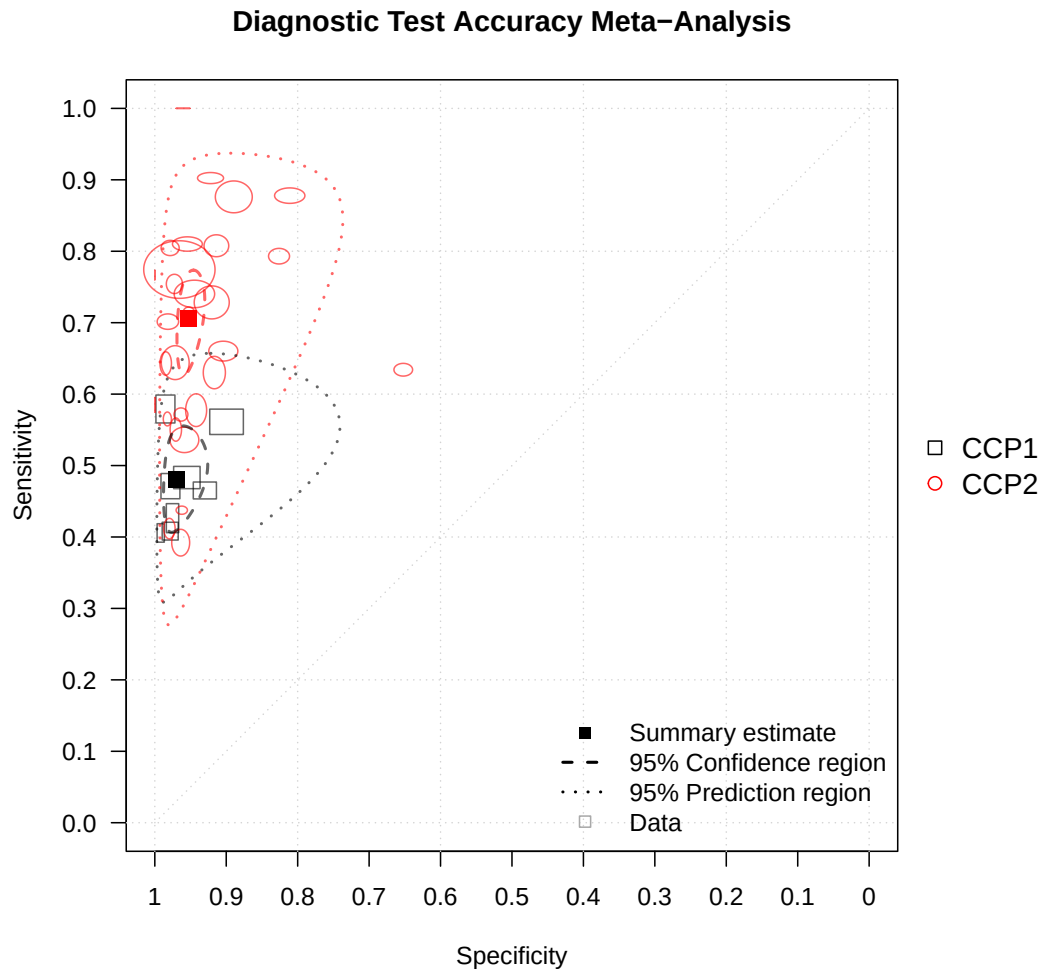
We can perform a likelihood ratio test via `anova`, essentially testing whether there exist subgroup differences in sensitivity.

```
anova(constrainsens,reitsmasub)
#>           Df logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1  6 -272.77
#> Model 2  7 -266.69      1 12.181 0.0004827 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do I allow for different subgroup-specific random-effects (co-)variances in the Reitsma model?

```
heteroskedastic <- fitReitsmaSubgroup(data=anticcp,
                                     TP=TP,
                                     FP=FP,
                                     FN=FN,
                                     TN=TN,
                                     study=study,
                                     subgroup=generation,
                                     variances="unequal")
plot(heteroskedastic,
     scale=0.01,
     nudge_legend=-0.2,
```

```
size="se",
col=c("black","red"))
```



```
anova(reitsmasub,heteroskedastic)
#>           Df logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1   7 -266.69
#> Model 2  10 -262.42      3  8.5306    0.03623 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do I fit the Reitsma subgroup model using the custom TMB backend?

By default, `fitReitsmaSubgroup()` fits the Reitsma model using `glmmTMB`. The package also provides `fitReitsmaSubgroupTMB()`, which fits the same binomial-normal model using a custom Template Model Builder (TMB) template.

The custom TMB backend may offer improved numerical robustness in difficult or near-boundary cases.

```
data("anticcp")
reitsmasubTMB <- fitReitsmaSubgroupTMB(data=anticcp,
                                       TP=TP,
                                       FP=FP,
                                       FN=FN,
                                       TN=TN,
                                       study=study,
                                       variances="unequal",
                                       subgroup=generation)

reitsmasubTMB
#>
#> Reitsma Subgroup TMB Model
#> -----
#>
#> Number of studies      : 37
#> Number of subgroups   : 2
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 7.201756e-05
#> -2 log likelihood      : 524.839 ( df = 10 )
#> AIC                   : 544.839
#> BIC                   : 560.948
#>
#> Use summary() for parameter estimates.
summary(reitsmasubTMB)
#>
#> Summary: Reitsma Subgroup TMB Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate Std. Error
#> mu_A.CCP1      -0.077734976 0.093745405
#> mu_A.CCP2       0.872414330 0.137541560
```

```

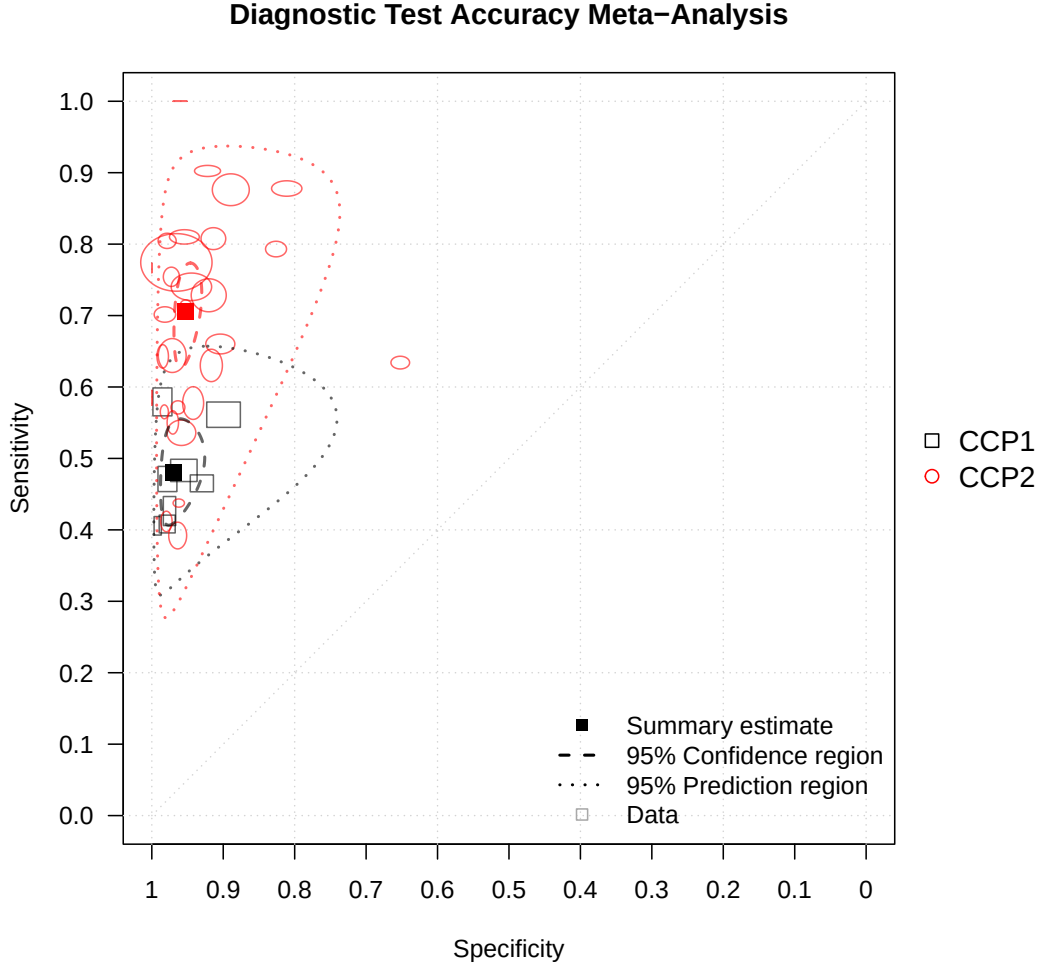
#> mu_B.CCP1      3.458614000 0.290964660
#> mu_B.CCP2      3.014784100 0.164919640
#> sigma2_A.CCP1  0.042771854 0.033050876
#> sigma2_A.CCP2  0.481452150 0.152370450
#> sigma2_B.CCP1  0.479322870 0.320124020
#> sigma2_B.CCP2  0.560479850 0.215712700
#> sigma_AB.CCP1 -0.062584493 0.079018858
#> sigma_AB.CCP2 -0.233390560 0.129708840
#> rho_AB.CCP1    -0.437092770 0.445469900
#> rho_AB.CCP2    -0.449289960 0.200921540
#> nu_A.CCP2      0.950149300 0.166450840
#> nu_B.CCP2      -0.443829930 0.334453170
#>
#> Sensitivity / specificity
#> -----
#>           type   Estimate conflevel   CI_Lower   CI_Upper
#> mu_A.CCP1 sens 0.48057604      0.95 0.43500175 0.52647587
#> mu_A.CCP2 sens 0.70524782      0.95 0.64630529 0.75804497
#> mu_B.CCP1 spec 0.96948699      0.95 0.94726671 0.98251710
#> mu_B.CCP2 spec 0.95323757      0.95 0.93652604 0.96571036
#>
#> Recovered HSRROC parameters (Rutter-Gatsonis)
#> -----
#>           Lambda      Theta      beta sigma2_alpha sigma2_theta
#> CCP1 1.7480891 -1.01627220 1.208247100 0.16119811 0.10288402
#> CCP2 3.8085824 -0.99809026 0.075993224 0.57214955 0.37642794
#>
#> Subgroups
#> -----
#> [1] "CCP1" "CCP2"

```

```

plot(reitsmasubTMB,
     scale=0.01,
     nudge_legend=-0.2,
     size="se",
     col=c("black","red"))

```



How do we include covariates in the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\text{logit}(\pi_{ij}) = (z_i^\top \theta_i + z_i^\top \alpha_i x_{ij}) \exp(-z_i^\top \beta x_{ij}),$$

$$z_i^\top \alpha_i \sim \mathcal{N}(z_i^\top \Lambda, \sigma_\alpha^2), \quad z_i^\top \theta_i \sim \mathcal{N}(z_i^\top \Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

Let's assume that z_i includes a single categorical covariate representing three subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \theta_i = \begin{cases} \theta_i & \text{for subgroup 1,} \\ \theta_i + \gamma_2 & \text{for subgroup 2,} \\ \theta_i + \gamma_3 & \text{for subgroup 3,} \end{cases} \quad z_i^\top \alpha_i = \begin{cases} \alpha_i & \text{for subgroup 1,} \\ \alpha_i + \xi_2 & \text{for subgroup 2,} \\ \alpha_i + \xi_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \beta = \begin{cases} \beta & \text{for subgroup 1,} \\ \beta + \delta_2 & \text{for subgroup 2,} \\ \beta + \delta_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \Lambda = \begin{cases} \Lambda & \text{for subgroup 1,} \\ \Lambda + \xi_2 & \text{for subgroup 2,} \\ \Lambda + \xi_3 & \text{for subgroup 3,} \end{cases} \quad z_i^\top \Theta_i = \begin{cases} \Theta & \text{for subgroup 1,} \\ \Theta + \gamma_2 & \text{for subgroup 2,} \\ \Theta + \gamma_3 & \text{for subgroup 3.} \end{cases}$$

Note: For prediction `fitRutterGatsonisSubgroup()` uses the prediction matrix

$$Z_{\text{pred}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

for the case above and therefore recovers the threshold, accuracy, and shape parameters for each subgroup as if it were the reference group.

How do I perform a Rutter and Gatsonis meta-regression with a single categorical study-level covariate?

```
data("RF")
RF2      <- RF[RF$method %in% c("LA","ELISA","Nephelometry"),]
RF2$method <- factor(RF2$method,levels=c("LA","ELISA","Nephelometry"))
ruttergatsonissub <- fitRutterGatsonisSubgroup(data=RF2,
                                              TP=TP,
                                              FP=FP,
```

```

FN=FN,
TN=TN,
study=study,
subgroup=method,
constrain="shape") # assumes equal
                    # shapes in subgroups

```

```

ruttergatsonissub
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups   : 3
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 0.0002867077
#> -2 log likelihood      : 753.722 ( df = 9 )
#> AIC                   : 771.722
#> BIC                   : 788.374
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissub)
#>
#> Summary: Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate Std. Error
#> Lambda_LA      2.456309800 0.323577140
#> xi_ELISA        0.248533990 0.439541130
#> xi_Nephelometry 0.331428920 0.442600640
#> Theta_LA       -0.550060260 0.213340320
#> gamma_ELISA     -0.196255960 0.260961790
#> gamma_Nephelometry 0.495863020 0.262230320
#> beta_LA         0.197677370 0.169839730
#> delta_ELISA     0.000000000 0.000000000
#> delta_Nephelometry 0.000000000 0.000000000
#> sigma2_alpha    1.277914900 0.308524820
#> sigma2_theta    0.476839640 0.113444950
#> Lambda_LA      2.456309800 0.323577140

```

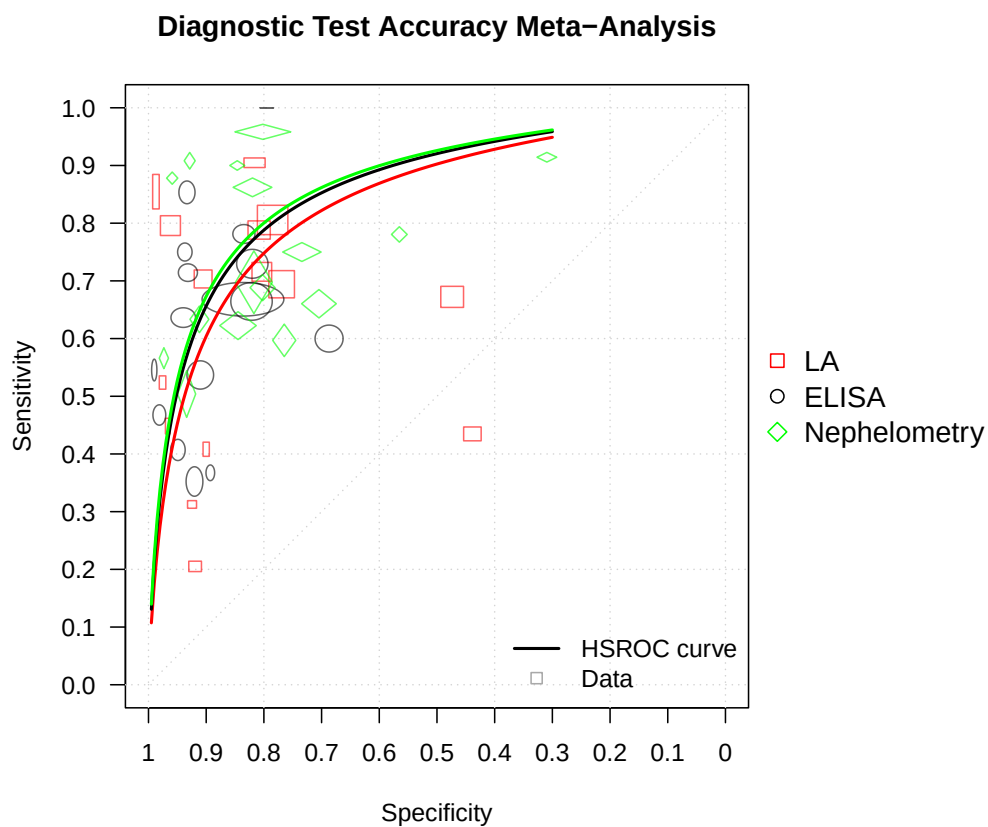
```

#> Lambda_ELISA      2.704843800 0.326954560
#> Lambda_Nephelometry 2.787738800 0.305779190
#> Theta_LA          -0.550060260 0.213340320
#> Theta_ELISA       -0.746316220 0.209913080
#> Theta_Nephelometry -0.054197236 0.212131240
#> beta_LA           0.197677370 0.169839730
#> beta_ELISA        0.197677370 0.169839730
#> beta_Nephelometry 0.197677370 0.169839730
#> logitsens         0.826168420 0.286300410
#> logitsens         1.051312600 0.286070100
#> logitsens         1.126406200 0.276895840
#> sens              0.695544150 0.060627687
#> sens              0.741026870 0.054898582
#> sens              0.755175060 0.051194078
#>
#> Sensitivity / specificity
#> -----
#>      subgroup      spec conflevel logitsens Std_Error CI_Lower CI_Upper
#> 1      LA 0.84615385      0.95 0.82616842 0.28630041 0.26502992 1.3873069
#> 2      ELISA 0.84615385      0.95 1.05131260 0.28607010 0.49062549 1.6119997
#> 3 Nephelometry 0.84615385      0.95 1.12640620 0.27689584 0.58370030 1.6691121
#>      Sens SensCI_Lower SensCI_Upper
#> 1 0.69554415 0.56587235 0.80016196
#> 2 0.74102687 0.62025377 0.83368883
#> 3 0.75517506 0.64191840 0.84145740
#>
#> Recovered bivariate parameters (Reitsma)
#> -----
#>      mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec sigma_AB
#> LA      0.61427837 1.9629508 0.65348637 0.97036902 -0.15736092
#> ELISA    0.54906437 2.3167723 0.65348637 0.97036902 -0.15736092
#> Nephelometry 1.21359400 1.5985037 0.65348637 0.97036902 -0.15736092
#>
#> Subgroups
#> -----
#> [1] "LA" "ELISA" "Nephelometry"

```

How do I get a summary plot of the Rutter and Gatsonis subgroup model?

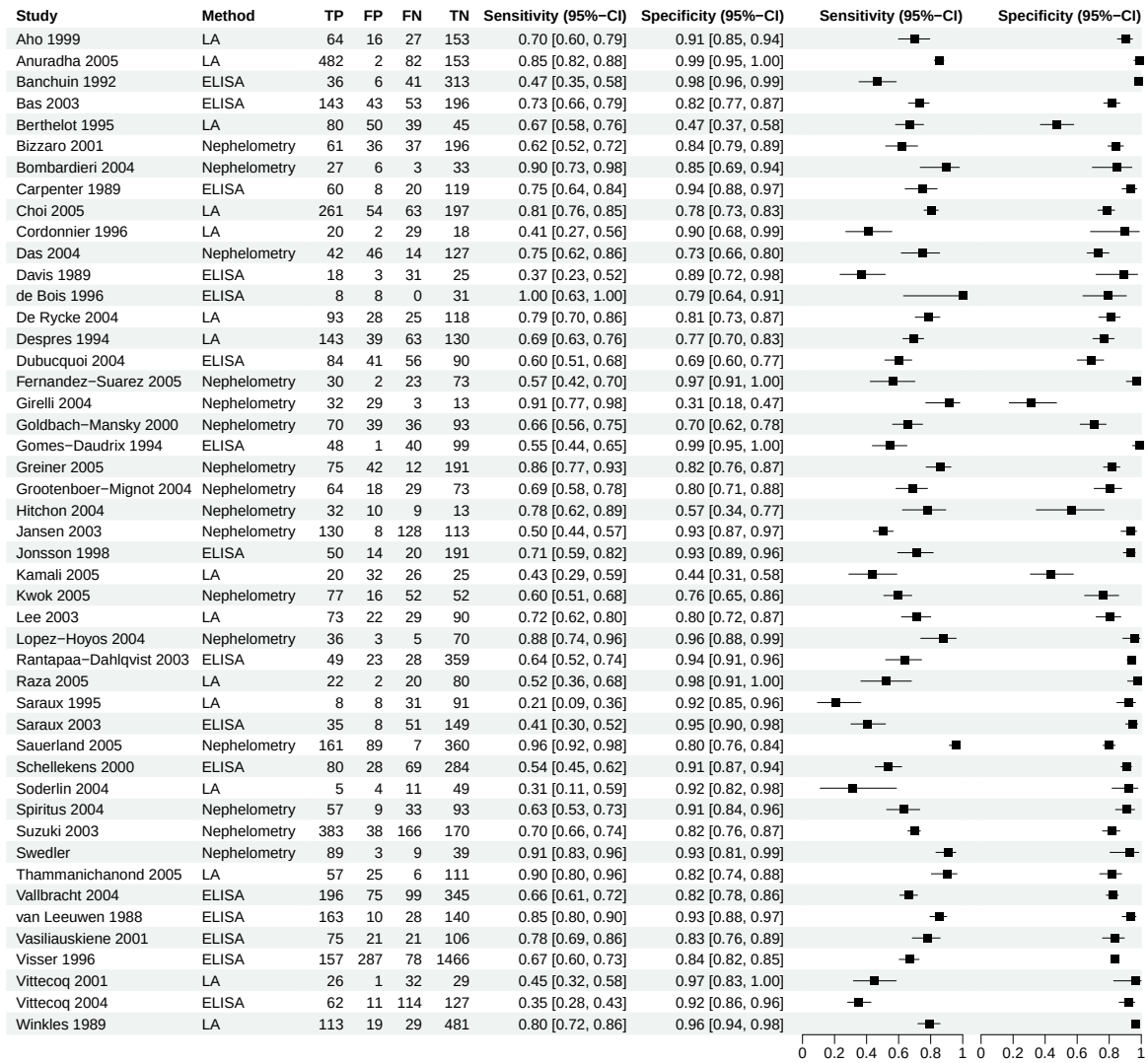
```
plot(ruttergatsonissub,
     specrange=c(0.3,0.995),
     size="se",
     col=c("red","black","green"),
     scale=0.015)
```



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(ruttergatsonissub, subgroup_label = "Method")
```



How do I constrain parameters in the Rutter and Gatsonis subgroup model?

In the Rutter and Gatsonis model, all parameter constraints are controlled by `constrain`, i.e.,

- `constrain="sigma2_alpha"` sets σ_{α}^2 to zero,
- `constrain="sigma2_theta"` sets σ_{θ}^2 to zero,

- `constrain="accuracy"` assumes equal accuracy parameters across subgroups, i.e. $\xi_2 = \xi_3 = \dots = 0$,
- `constrain="threshold"` assumes equal threshold parameters across subgroups, i.e. $\gamma_2 = \gamma_3 = \dots = 0$,
- `constrain="shape"` assumes equal shape parameters across subgroups, i.e. $\delta_2 = \delta_3 = \dots = 0$,
- `constrain="shape_zero"` fixes all shape parameters at zero.

Constraints can also be combined, for example `constrain=c("shape","sigma2_theta")`.

How do I use the general Rutter and Gatsonis regression function?

This method is for advanced users who feel comfortable specifying their own design and prediction matrices. For study-level covariates, the design matrix Z needs two identical consecutive rows per study, one for the diseased and one for the non-diseased. Of note, there are neither `plot()` nor `forest()` methods for `fitRutterGatsonisReg()`.

Let's reproduce the subgroup-analysis from before.

```
# Specify design matrix Z
Z <- model.matrix(~method,data=RF2)
# For study level-covariates, we need to two identical consecutive
# rows per study (diseased and non-diseased).
Z2 <- Z[rep(seq_len(nrow(Z)), each = 2), , drop = FALSE]
# Specify prediction matrix Z_pred
Z_pred <- matrix(c(1,0,0,1,1,0,1,0,1),ncol=3,nrow=3,byrow=T)
constrain <- list(shape_coef=factor(c(1, rep(NA, ncol(Z2) - 1))))

ruttergatsonisreg <- fitRutterGatsonisReg(data=RF2,
                                         TP=TP,
                                         FP=FP,
                                         FN=FN,
                                         TN=TN,
                                         study=study,
                                         Z=Z2,
                                         Z_pred=Z_pred,
                                         map=constrain)

ruttergatsonisreg
#>
#> Rutter & Gatsonis Regression Model
#> -----
#>
```

```

#> Number of studies : 47
#> Optimizer          : Converged
#> Hessian            : Positive definite
#> Max |grad|         : 0.0003981125
#> -2 log likelihood : 753.722 ( df = 9 )
#> AIC                : 771.722
#> BIC                : 788.374
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonisreg)
#>
#> Summary: Rutter & Gatsonis Regression Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate Std. Error
#> accuracy_coef  2.456306200 0.323578110
#> accuracy_coef  0.248536610 0.439542730
#> accuracy_coef  0.331419970 0.442602170
#> threshold_coef -0.550050810 0.213340320
#> threshold_coef -0.196254760 0.260962000
#> threshold_coef  0.495866140 0.262230550
#> shape_coef     0.197682730 0.169839700
#> shape_coef     0.000000000 0.000000000
#> shape_coef     0.000000000 0.000000000
#> sigma2_alpha   1.277925600 0.308528070
#> sigma2_theta   0.476840430 0.113445210
#> Lambda_Pred    2.456306200 0.323578110
#> Lambda_Pred    2.704842800 0.326955320
#> Lambda_Pred    2.787726100 0.305780160
#> Theta_Pred     -0.550050810 0.213340320
#> Theta_Pred     -0.746305570 0.209913120
#> Theta_Pred     -0.054184665 0.212131320
#> beta_Pred      0.197682730 0.169839700
#> beta_Pred      0.197682730 0.169839700
#> beta_Pred      0.197682730 0.169839700
#> logitsens      0.826166620 0.286300670
#> logitsens      1.051312600 0.286070290
#> logitsens      1.126395500 0.276896050
#> sens           0.695543770 0.060627783
#> sens           0.741026870 0.054898619

```

```

#> sens          0.755173080 0.051194397
#>
#> Sensitivity / specificity
#> -----
#>          spec conflevel  logitsens  Std_Error  CI_Lower  CI_Upper      Sens
#> 1 0.84615385      0.95 0.82616662 0.28630067 0.26502762 1.3873056 0.69554377
#> 2 0.84615385      0.95 1.05131260 0.28607029 0.49062509 1.6120000 0.74102687
#> 3 0.84615385      0.95 1.12639550 0.27689605 0.58368917 1.6691018 0.75517308
#>  SensCI_Lower SensCI_Upper
#> 1    0.56587179  0.80016175
#> 2    0.62025368  0.83368888
#> 3    0.64191584  0.84145602

```

How do I compare models?

In the previous section we fitted the RF data set using the Rutter and Gatsonis subgroup model while keeping the shape parameter equal across all subgroups `constrain="shape"`. Now let's fit the full model allowing for different shape parameters across subgroups and check whether the data lend support to this approach. In addition, let's also consider a model that constrains both `shape` and `accuracy`.

```

ruttergatsonissubfull <- fitRutterGatsonisSubgroup(data=RF2,
                                                    TP=TP,
                                                    FP=FP,
                                                    FN=FN,
                                                    TN=TN,
                                                    study=study,
                                                    subgroup=method,
                                                    constrain=NULL)

ruttergatsonissubfull
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups   : 3
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 0.0001936155
#> -2 log likelihood      : 753.553 ( df = 11 )
#> AIC                   : 775.553

```

```

#> BIC : 795.905
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissubfull)
#>
#> Summary: Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#> Estimate Std. Error
#> Lambda_LA 2.42435650 0.330490060
#> xi_ELISA 0.29743443 0.513150050
#> xi_Nephelometry 0.37161351 0.450864220
#> Theta_LA -0.50294636 0.244516610
#> gamma_ELISA -0.25783318 0.370244910
#> gamma_Nephelometry 0.39706424 0.359443140
#> beta_LA 0.27775702 0.266422660
#> delta_ELISA -0.10290405 0.426607590
#> delta_Nephelometry -0.16039260 0.397537120
#> sigma2_alpha 1.27305060 0.307984420
#> sigma2_theta 0.47624252 0.113436600
#> Lambda_LA 2.42435650 0.330490060
#> Lambda_ELISA 2.72179090 0.392992730
#> Lambda_Nephelometry 2.79597000 0.307080030
#> Theta_LA -0.50294636 0.244516610
#> Theta_ELISA -0.76077954 0.278169440
#> Theta_Nephelometry -0.10588212 0.263228630
#> beta_LA 0.27775702 0.266422660
#> beta_ELISA 0.17485297 0.333210110
#> beta_Nephelometry 0.11736442 0.295002910
#> logitsens 0.81868527 0.275188790
#> logitsens 1.06266640 0.324053890
#> logitsens 1.12065150 0.288348430
#> sens 0.69395719 0.058444763
#> sens 0.74319976 0.061846941
#> sens 0.75410954 0.053467971
#>
#> Sensitivity / specificity
#> -----
#>
#> subgroup spec conflevel logitsens Std_Error CI_Lower CI_Upper
#> 1 LA 0.84615385 0.95 0.81868527 0.27518879 0.27932515 1.3580454

```

```

#> 2      ELISA 0.84615385      0.95 1.06266640 0.32405389 0.42753242 1.6978003
#> 3 Nephelometry 0.84615385      0.95 1.12065150 0.28834843 0.55549896 1.6858041
#>      Sens SensCI_Lower SensCI_Upper
#> 1 0.69395719 0.56938077 0.79544184
#> 2 0.74319976 0.60528428 0.84524723
#> 3 0.75410954 0.63541045 0.84367155
#>
#> Recovered bivariate parameters (Reitsma)
#> -----
#>      mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec      sigma_AB
#> LA      0.61726846 1.9706516      0.60182246      1.04887820 -0.15797987
#> ELISA    0.54987796 2.3155155      0.66705102      0.94631215 -0.15797987
#> Nephelometry 1.21846130 1.5947581      0.70652253      0.89344424 -0.15797987
#>
#> Subgroups
#> -----
#> [1] "LA"      "ELISA"      "Nephelometry"
ruttergatsonissubacc <- fitRutterGatsonisSubgroup(data=RF2,
                                                    TP=TP,
                                                    FP=FP,
                                                    FN=FN,
                                                    TN=TN,
                                                    study=study,
                                                    subgroup=method,
                                                    constrain=c("shape", "accuracy"))

ruttergatsonissubacc
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups   : 3
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 0.0005592484
#> -2 log likelihood      : 754.332 ( df = 7 )
#> AIC                   : 768.332
#> BIC                   : 781.283
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissubacc)
#>

```

```

#> Summary: Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>               Estimate Std. Error
#> Lambda_LA      2.65810190 0.191815720
#> xi_ELISA        0.00000000 0.000000000
#> xi_Nephelometry 0.00000000 0.000000000
#> Theta_LA       -0.55746273 0.212932400
#> gamma_ELISA    -0.19386257 0.260935930
#> gamma_Nephelometry 0.49703130 0.262130440
#> beta_LA        0.18948853 0.164729720
#> delta_ELISA     0.00000000 0.000000000
#> delta_Nephelometry 0.00000000 0.000000000
#> sigma2_alpha    1.29063570 0.312335810
#> sigma2_theta    0.47671067 0.113444250
#> Lambda_LA      2.65810190 0.191815720
#> Lambda_ELISA    2.65810190 0.191815720
#> Lambda_Nephelometry 2.65810190 0.191815720
#> Theta_LA       -0.55746273 0.212932400
#> Theta_ELISA    -0.75132530 0.208099240
#> Theta_Nephelometry -0.06043143 0.209582940
#> beta_LA        0.18948853 0.164729720
#> beta_ELISA     0.18948853 0.164729720
#> beta_Nephelometry 0.18948853 0.164729720
#> logitsens      1.00734590 0.165264820
#> logitsens      1.00734590 0.165264820
#> logitsens      1.00734590 0.165264820
#> sens           0.73250042 0.032382577
#> sens           0.73250042 0.032382577
#> sens           0.73250042 0.032382577
#>
#> Sensitivity / specificity
#> -----
#>
#>      subgroup      spec conflevel logitsens Std_Error CI_Lower CI_Upper
#> 1      LA 0.84615385      0.95 1.0073459 0.16526482 0.68343284 1.331259
#> 2      ELISA 0.84615385      0.95 1.0073459 0.16526482 0.68343284 1.331259
#> 3 Nephelometry 0.84615385      0.95 1.0073459 0.16526482 0.68343284 1.331259
#>
#>      Sens SensCI_Lower SensCI_Upper
#> 1 0.73250042 0.66450444 0.79104882
#> 2 0.73250042 0.66450444 0.79104882

```

```

#> 3 0.73250042 0.66450444 0.79104882
#>
#> Recovered bivariate parameters (Reitsma)
#> -----
#>          mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec  sigma_AB
#> LA          0.70184091 2.0739910 0.66138418 0.96614309 -0.15405174
#> ELISA        0.52550245 2.2871192 0.66138418 0.96614309 -0.15405174
#> Nephelometry 1.15394330 1.5275659 0.66138418 0.96614309 -0.15405174
#>
#> Subgroups
#> -----
#> [1] "LA"          "ELISA"          "Nephelometry"

```

How do I get the log likelihood, the AIC, and the BIC of a model?

```

logLik(ruttergatsonissubfull)
#> 'log Lik.' -376.7765 (df=11)
AIC(ruttergatsonissubfull)
#> [1] 775.5531
BIC(ruttergatsonissubfull)
#> [1] 795.9047

logLik(ruttergatsonissub)
#> 'log Lik.' -376.8612 (df=9)
AIC(ruttergatsonissub)
#> [1] 771.7223
BIC(ruttergatsonissub)
#> [1] 788.3736

logLik(ruttergatsonissubacc)
#> 'log Lik.' -377.1659 (df=7)
AIC(ruttergatsonissubacc)
#> [1] 768.3318
BIC(ruttergatsonissubacc)
#> [1] 781.2829

```

How do I perform a likelihood ratio test of the two models?

The tests below formally investigate whether the HSROC curve shapes and/or accuracy parameters are equal in all subgroups.

```
anova(ruttergatsonissub,
      ruttergatsonissubfull)
#>           Df logLik Df.diff Chisq Pr(>Chisq)
#> Model 1   9 -376.86
#> Model 2  11 -376.78      2 0.1692      0.9189
anova(ruttergatsonissubacc,
      ruttergatsonissub)
#>           Df logLik Df.diff Chisq Pr(>Chisq)
#> Model 1   7 -377.17
#> Model 2   9 -376.86      2 0.6095      0.7373
anova(ruttergatsonissubacc,
      ruttergatsonissubfull)
#>           Df logLik Df.diff Chisq Pr(>Chisq)
#> Model 1   7 -377.17
#> Model 2  11 -376.78      4 0.7787      0.9413
```

How do I replicate results from the Cochrane Handbook with respect to the schuetz data set?

What are the results for the full data set?

```
data(schuetz)
head(schuetz)
#>   test      study TP FP FN TN indirect
#> 1  CT Achenbach 2005 25  4  0 19      1
#> 2  CT Alkadhi 2008 57 12  2 79      1
#> 3  CT Andreini 2007 17  0  0 44      1
#> 4  CT Bayrak 2008 64  4  0 32      1
#> 5 MRI Bedaux 2002  7  1  0  1      1
#> 6 MRI Bogaert 2003 12  3  3  1      1
schuetz$test <- factor(schuetz$test, levels=c("MRI", "CT"))
schuetzreitsma <- fitReitsmaSubgroup(data=schuetz,
                                     TP=TP, FP=FP, FN=FN, TN=TN,
                                     study=study,
                                     subgroup=test,
                                     variances="common")

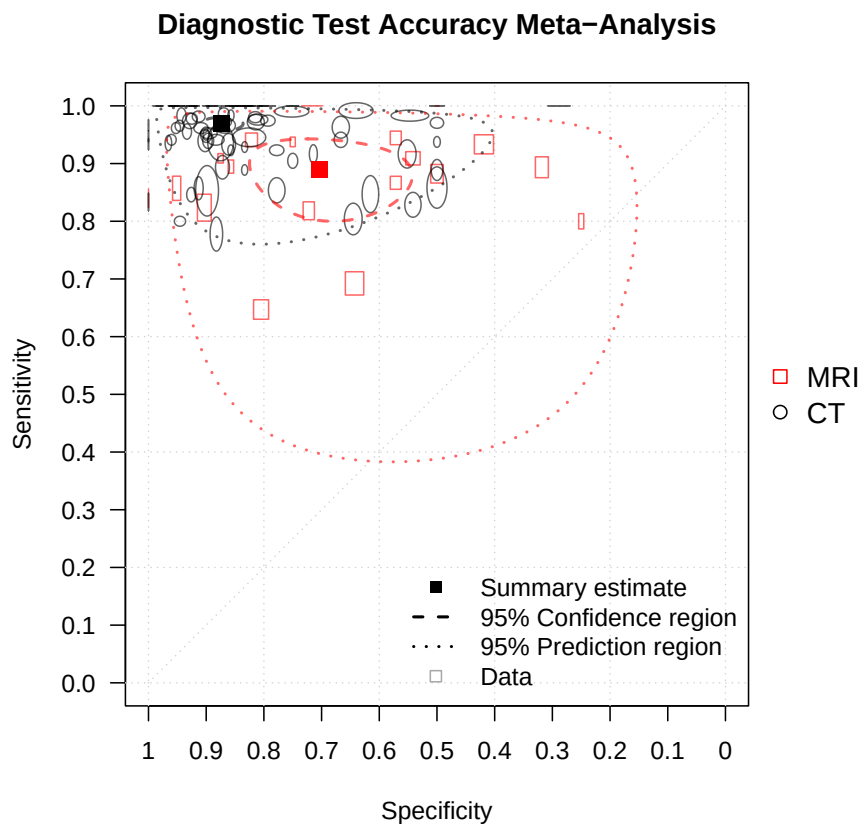
schuetzreitsma
#>
#> Reitsma Subgroup Model
#> -----
```

```

#>
#> Number of studies      : 108
#> Number of subgroups   : 2
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 0.0009488887
#> -2 log likelihood      : 951.843 ( df = 7 )
#> AIC                   : 965.843
#> BIC                   : 984.618
#>
#> Use summary() for parameter estimates.
summary(schuetzreitsma)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>
#> Estimate Std_Error
#> mu_A.MRI      2.09345490 0.26392178
#> mu_B.MRI      0.86047288 0.25722689
#> mu_A.CT       3.48272650 0.15698592
#> mu_B.CT       1.92906950 0.12066660
#> sigma2_A.sens.log_sigma_A 0.85004519 0.21954718
#> sigma2_B.spec.log_sigma_B 0.85268704 0.16833879
#> sigma_AB      0.18575027 0.13420795
#> nu_A.CT       1.38928530 0.30152230
#> nu_B.CT       1.06858890 0.28340397
#>
#> Sensitivity / specificity
#> -----
#>
#> type Estimate conflevel CI_Lower CI_Upper
#> mu_A.MRI sens 0.89026540 0.95 0.82866250 0.93154909
#> mu_B.MRI spec 0.70275944 0.95 0.58814870 0.79651089
#> mu_A.CT sens 0.97019227 0.95 0.95988410 0.97791261
#> mu_B.CT spec 0.87314639 0.95 0.84456148 0.89711495
#>
#> Recovered HSROC parameters (Rutter-Gatsonis)
#> -----
#>
#> Lambda Theta beta sigma2_alpha sigma2_theta
#> MRI 2.9548851 0.61763696 0.001551535 2.0742307 0.33280741
#> CT 5.4130029 0.77892791 0.001551535 2.0742307 0.33280741

```

```
#>
#> Subgroups
#> -----
#> [1] "MRI" "CT"
plot(schuetzreitsma,
     nudge_legend=-0.2,
     size="se",
     col=c("red","black"))
```



```
schuetzreitsma1 <- fitReitsmaSubgroup(data=schuetz,
                                     TP=TP,FP=FP,FN=FN,TN=TN,
                                     study=study,
                                     subgroup=test,
                                     variances="unequal")

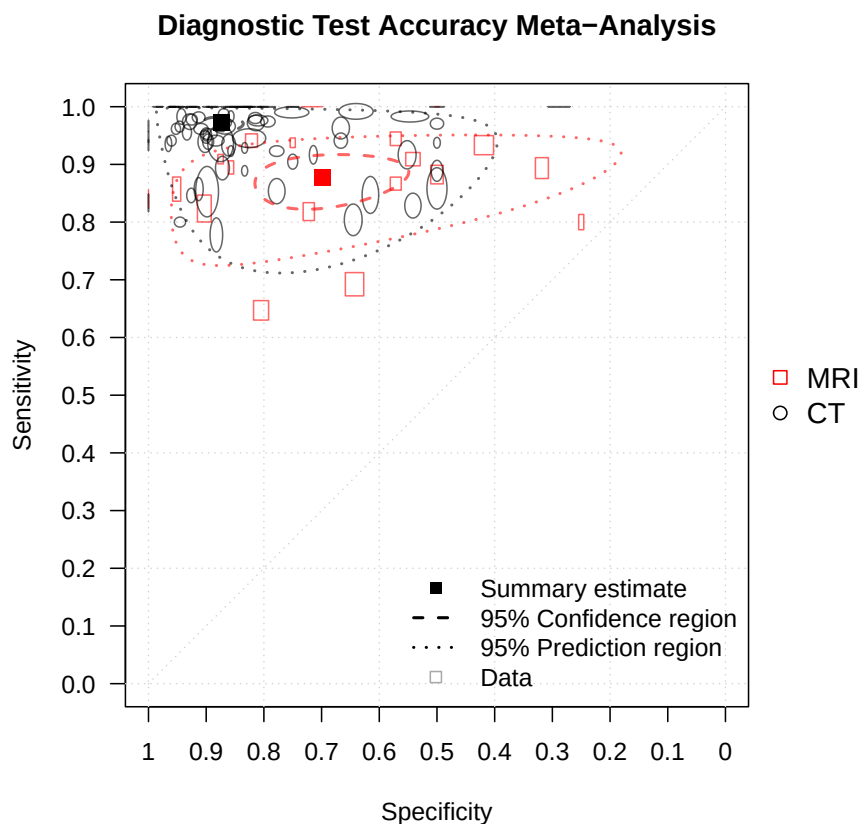
schuetzreitsma1
#>
```

```

#> Reitsma Subgroup Model
#> -----
#>
#> Number of studies      : 108
#> Number of subgroups   : 2
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 0.0001117501
#> -2 log likelihood      : 943.305 ( df = 10 )
#> AIC                   : 963.305
#> BIC                   : 990.127
#>
#> Use summary() for parameter estimates.
summary(schuetzreitsma1)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>           Estimate Std_Error
#> mu_A.MRI      1.96704120 0.16243413
#> mu_B.MRI      0.84102032 0.24063701
#> sigma2_A.MRI  0.11264202 0.16001562
#> sigma2_B.MRI  0.71212692 0.34247434
#> sigma_AB.MRI -0.14620818 0.13705823
#> mu_A.CT       3.55943060 0.17395078
#> mu_B.CT       1.93164220 0.12246545
#> sigma2_A.CT   1.10692640 0.30999078
#> sigma2_B.CT   0.88004569 0.19038875
#> sigma_AB.CT   0.30489374 0.17465399
#> nu_A.CT       1.59239900 0.23799792
#> nu_B.CT       1.09062870 0.27000566
#>
#> Sensitivity / specificity
#> -----
#>
#>           type  Estimate conflevel  CI_Lower  CI_Upper
#> mu_A.MRI sens 0.87729295      0.95 0.83871205 0.90766116
#> mu_B.MRI spec 0.69868006      0.95 0.59130927 0.78795796
#> mu_A.CT  sens 0.97233226      0.95 0.96152431 0.98016685
#> mu_B.CT  spec 0.87343107      0.95 0.84443634 0.89767647
#>

```

```
#> Recovered HSRROC parameters (Rutter-Gatsonis)
#> -----
#>      Lambda      Theta      beta sigma2_alpha sigma2_theta
#> MRI 3.6494747 1.29435110 0.92202065 0.27403018 0.21471572
#> CT 5.4067108 0.65771141 -0.11468430 2.58376400 0.34104725
#>
#> Subgroups
#> -----
#> [1] "MRI" "CT"
plot(schuetzreitsma1,
     nudge_legend=-0.2,
     size="se",
     col=c("red","black"))
```



```
schuetzreitsma2 <- fitReitsmaSubgroup(data=schuetz,
                                     TP=TP,FP=FP,FN=FN,TN=TN,
```

```

                                study=study,
                                subgroup=test,
                                sensspec_constrain="sens")
schuetzreitsma3 <- fitReitsmaSubgroup(data=schuetz,
                                TP=TP,FP=FP,FN=FN,TN=TN,
                                study=study,
                                subgroup=test,
                                sensspec_constrain="spec")

anova(schuetzreitsma,schuetzreitsma1)
#>           Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1    7 -475.92
#> Model 2   10 -471.65      3 8.5381    0.03611 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
anova(schuetzreitsma2,schuetzreitsma)
#>           Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1    6 -485.79
#> Model 2    7 -475.92      1 19.745 8.848e-06 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
anova(schuetzreitsma3,schuetzreitsma)
#>           Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1    6 -482.64
#> Model 2    7 -475.92      1 13.432 0.0002474 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

What are the results for the direct comparisons?

```

schuetz2 <- subset(schuetz,indirect==0)
schuetzreitsma4 <- fitReitsmaSubgroup(data=schuetz2,
                                TP=TP,FP=FP,FN=FN,TN=TN,
                                study=study,
                                subgroup=test,
                                constrain="sigma2_A")

schuetzreitsma4
#>
#> Reitsma Subgroup Model
#> -----
#>

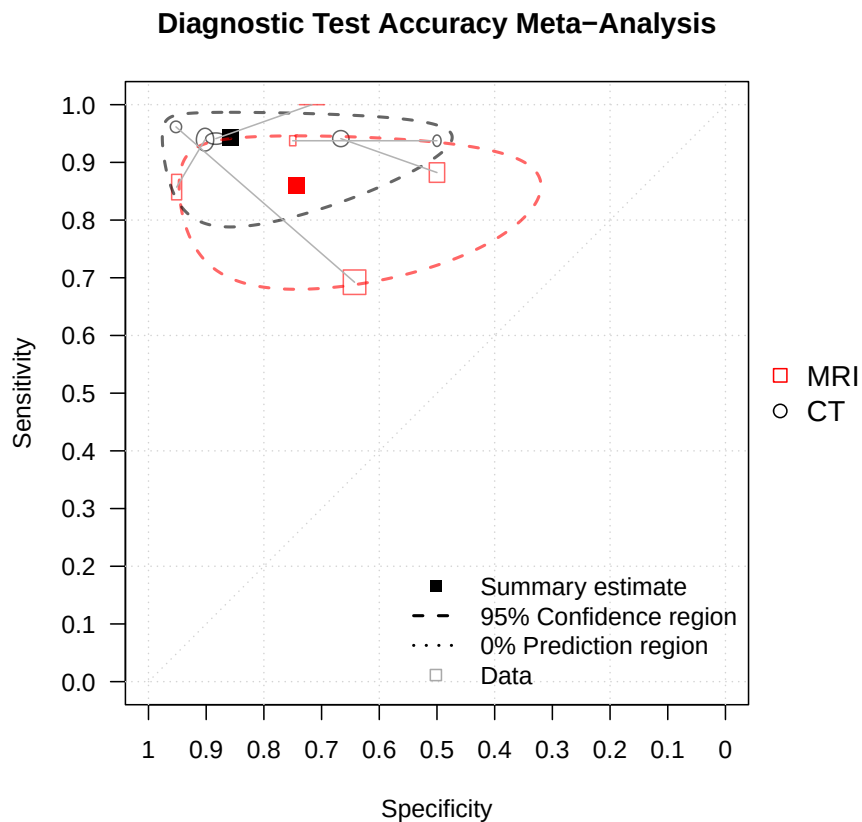
```

```

#> Number of studies      : 10
#> Number of subgroups   : 2
#> Optimizer              : Converged
#> Hessian                : Positive definite
#> Max |grad|             : 1.866797e-05
#> -2 log likelihood      : 88.734 ( df = 5 )
#> AIC                   : 98.734
#> BIC                   : 100.247
#>
#> Use summary() for parameter estimates.
summary(schuetzreitsma4)
#>
#> Summary: Reitsma Subgroup Model
#> -----
#>
#> Parameter estimates
#> -----
#>
#>
#> Estimate Std_Error
#> mu_A.MRI      1.8082893e+00 0.24124001
#> mu_B.MRI      1.0631770e+00 0.41604406
#> mu_A.CT       2.8134122e+00 0.34318787
#> mu_B.CT       1.8013974e+00 0.43628527
#> sigma2_A.sens.log_sigma_A 4.9303807e-32 0.00000000
#> sigma2_B.spec.log_sigma_B 5.8153371e-01 0.43286772
#> sigma_AB      0.0000000e+00 0.00000000
#> nu_A.CT       1.0051219e+00 0.41949313
#> nu_B.CT       7.3821676e-01 0.60559889
#>
#> Sensitivity / specificity
#> -----
#>
#> type Estimate conflevel CI_Lower CI_Upper
#> mu_A.MRI sens 0.85915500      0.95 0.79174360 0.90730054
#> mu_B.MRI spec 0.74329721      0.95 0.56162156 0.86745121
#> mu_A.CT sens 0.94339630      0.95 0.89480371 0.97028818
#> mu_B.CT spec 0.85831895      0.95 0.72036925 0.93441053
#>
#> Recovered HSROC parameters (Rutter-Gatsonis)
#> -----
#>
#> Lambda Theta beta sigma2_alpha sigma2_theta
#> MRI 105972100 52986049 35.77261 3.3865514e-16 8.4663784e-17
#> CT 164875820 82437911 35.77261 3.3865514e-16 8.4663784e-17
#>

```

```
#> Subgroups
#> -----
#> [1] "MRI" "CT"
plot(schuetzreitsma4, predlevel=0.000001,
     nudge_legend=-0.2,
     size="se", scale=0.0025,
     connectstudies = TRUE,
     col=c("red", "black"))
```



References

- Reitsma, J. B., et al. (2005). Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *Journal of Clinical Epidemiology*, 58(10), 982–990.
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